

Shabari Vignesh

shabari.v@ajobguide.com || +1(408)-390-3347 || San Jose – CA, USA

Summary

As a **Data Engineer with 5+ years** of experience, I specialize in building and optimizing end-to-end data solutions that deliver actionable business intelligence. My expertise spans the entire data lifecycle, from designing robust, scalable pipelines using Apache Airflow, Databricks Workflow, and AWS Glue to ingest high-volume data from APIs and relational sources. I am proficient in managing modern data platforms like Snowflake, where I optimize DBT workflows with incremental models and strategic query rewrites to slash execution times and reduce costs. With a strong foundation in both SQL (SQL Server, MySQL) and NoSQL (MongoDB, HBase) systems, I ensure the delivery of accurate, business-ready data to power advanced analytics, machine learning models, and critical reporting.

Skills

Programming Language:	Python, R, Java, Scala, SQL
Big Data Ecosystem:	Hadoop, Hive, HDFS, HBase, Apache Airflow, Apache Kafka, Apache Spark, Apache Flink
Cloud Technologies:	AWS (EMR, EC2, S3, Athena, Glue, Elasticsearch, Lambda, DynamoDB, Redshift, QuickSight, Kinesis), Azure (Data Lake, Databricks, Data Storage, Data Factory, Azure App Service, Azure SQL Database, Azure Blob Storage), GCP, BigQuery
Visualizations:	Tableau, Power BI, Excel
ETL Tools:	SSIS, SSRS, Fivetran, SAS, Informatica, PySpark, Tableau, Talend
Packages & Data Processing:	Pandas, Matplotlib, Seaborn, PySpark, Data Pipelines, Data Build Tool (DBT)
Version Control & Database:	GitHub, Git, SQL Server, PostgreSQL, Cassandra, DynamoDB, MySQL, Snowflake
Data Management:	Data Modeling, Data Warehousing, Data Governance, Metadata Management, Data Quality, Master Data Management (MDM), Data Cataloging

Work Experience

JP Morgan Chase & Co., CAJanuary 2025 – Present

Data Engineer

- Engineered and managed high-throughput, distributed data processing systems using **Databricks, PySpark, and Python** to transform **5 TB of weekly raw transactional data** into curated datasets for risk analysis and regulatory reporting.
- Designed and automated mission-critical data pipelines with **Google Cloud Dataflow** and **Cloud Functions**, eliminating manual processes, saving **450+ person-hours annually**, and guaranteeing flawless data reconciliation for security and financial applications.
- Architected and optimized the **Snowflake** data cloud, refining data models through strategic clustering and pre-aggregated views to enhance analytical dashboard performance for **180+ financial partners**, slashing average query from **14 seconds to 5 seconds**.
- Established a robust, standardized infrastructure-as-code framework using **Terraform** and **GitLab CI/CD** to automate data platform deployments, reducing environment provisioning time from **3 days to under 8 hours** while enforcing stringent compliance and security controls.
- Operated and scaled real-time event streaming platforms with **Apache Kafka** and **Google Cloud Pub/Sub** to ingest and process over **2 million financial events monthly**, resulting in a **65% reduction in data pipeline** for critical security applications.
- Innovated an internal data exploration tool powered by advanced language models to interpret natural language queries, streamlining data retrieval and saving the analyst team **15-20 hours per month** on routine reporting tasks.

LTI Mindtree, IndiaSeptember 2021 – July 2023

Data Engineer / Machine Learning Engineer

- Engineered and managed large-scale, distributed ETL pipelines on AWS (EMR, S3) using **PySpark and SQL** within the Databricks environment, processing terabytes of multi-source data.
- Automated **data validation frameworks and operational reporting** using PySpark, reducing manual data handling by 30% and accelerating critical monthly business reporting cycles.
- Established end-to-end **MLOps pipelines** on **AWS (SageMaker, Lambda)** to automate model retraining, validation, and deployment, ensuring model reliability and cutting the average time-to-production for new algorithms from **3 weeks to 5 days**.
- Designed and productionized a fully automated, end-to-end **machine learning pipeline on Databricks** that manages data ingestion, feature engineering, exploratory analysis, and model selection, training hundreds of models per run and automatically deploying the best-performing model.
- Increased data availability for **business intelligence by 40%** by building robust and scalable data ingestion and processing frameworks.
- Performed deep-dive analysis on complex business data using **Python and SQL** to uncover trends and patterns, delivering insights that directly informed strategic product development and business planning.
- Collaborated with cross-functional team and marketing teams to **identify data infrastructure gaps** and deliver scalable data solutions, streamlining data workflows across the organization.
- Optimized predictive models for customer behaviour and value by applying hyperparameter tuning and **AutoML, resulting in a 15% increase in model accuracy** and more efficient budget allocation for customer retention programs.

Fusion Software Solutions, IndiaFebruary 2019 – August 2021

Data Engineer I

- Designed, developed, and managed scalable, efficient data pipelines on **Azure (Blob Storage, Data Factory) and Talend**, processing over **10 TB** of data daily. Optimized complex **ETL** workflows by implementing parallel processing, which reduced data processing time by **45%**.
- Engineered and maintained a high-throughput, low stream processing system using **Apache Flink for real-time analytics** on live data, enabling sub-second decision-making and enhancing operational intelligence.
- Upheld rigorous **data governance, quality, and compliance standards** by implementing automated data validation and cleansing routines. This initiative reduced data discrepancies by **over 90%**, ensured data reliability for reporting, and enforced strict data retention policies.
- Automated critical data validation and cleansing tasks by developing **custom Python scripts**, which slashed manual data handling effort by **40%** and improved overall pipeline data integrity.
- Collaborated with cross-functional business teams to translate data into actionable insights by developing interactive **Power BI dashboards**. Visualized key **KPIs** such as trends, directly contributing to improved customer retention strategies.
- Supported high-availability cluster management and job coordination in distributed environments by **configuring Apache Zookeeper**, significantly increasing system fault tolerance and reliability.

Education

Masters in Applied Data Science,
San Jose State University, San Jose, CA, USA.

Bachelors in Computer Science Engineering,
Visvesvaraya Technological University, India.